

Table 3. Preservation of medical and general QA performance

Method	Qwen2.5 7B					GPT-6 Luna					Gemma 4 12B					Qwen3.8 27B OFF					Qwen3.8 27B ON				
	MedQA	PubMedQA	Medbullets	MMLU	GSM8K	MedQA	PubMedQA	Medbullets	MMLU	GSM8K	MedQA	PubMedQA	Medbullets	MMLU	GSM8K	MedQA	PubMedQA	Medbullets	MMLU	GSM8K	MedQA	PubMedQA	Medbullets	MMLU	GSM8K
	Acc. ↑	Acc. ↑	Acc. ↑	Acc. ↑	EM ↑	Acc. ↑	Acc. ↑	Acc. ↑	Acc. ↑	EM ↑	Acc. ↑	Acc. ↑	Acc. ↑	Acc. ↑	EM ↑	Acc. ↑	Acc. ↑	Acc. ↑	Acc. ↑	EM ↑	Acc. ↑	Acc. ↑	Acc. ↑	Acc. ↑	EM ↑
Plain	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Balanced instruction	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CoT	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
PreWoMe	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Extract + Verify	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
TF-IDF text gate	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Direct gate	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CoT gate	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Probe gate	—	—	—	—	—	n/a	n/a	n/a	n/a	n/a	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
GEPA-gate	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
GEPA	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
GEPA (both stages)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Ours	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

No compatible completed QA measurements are available; all result cells remain blank.
Freeze the Table 2 response methods, prompts and gates before evaluation. Acc. = accuracy; EM = final-answer exact match.
GEPA-gate optimizes detection; GEPA optimizes direct answering; GEPA (both stages) optimizes detection and downstream answering.
Use identical questions and scoring rules within each model/dataset. Dash = unmeasured; n/a = native hidden probe unavailable.